



Departamento de Electrónica, Sistemas e Informática

Big Data

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Examples on Structured Streaming (files)

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Create SparkSession

```
In [1]: from pcamarillor.spark_utils import SparkUtils

su = SparkUtils("Structured Streaming with Files",
                master_url="spark://spark-master:7077")

su.spark
```

```
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/04/07 02:35:54 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
```

Out[1]: **SparkSession - in-memory**

SparkContext

Spark UI

Version	v4.0.1
Master	spark://spark-master:7077
AppName	Structured Streaming with Files

Create a data stream from a local socket

Connect Spark to the socket

```
In [3]: !mkdir -p /opt/spark/work-dir/data/streaming/logs/
```

```
In [5]: !ls /opt/spark/work-dir/data/streaming/logs/
```

```
In [ ]: import pyspark.sql.functions as F
        from pathlib import Path
        import shutil

        logs_schema = SparkUtils.generate_schema([("raw_line", "string")])

        input_path = "/opt/spark/work-dir/data/streaming/logs/"

        # Create the stream
        logs_df = (su.spark.readStream
                   .format("text")
```

```
        .option("maxFilesPerTrigger", 1) # Let's process one file at a time
        .schema(logs_schema)
        .load(input_path))

# Transform original dataframe
parsed_df = (
    logs_df
    .withColumn("parts", F.split(F.col("raw_line"), r" \| "))
    .withColumn("timestamp", F.to_timestamp(F.col("parts")[0], "yyyy-MM-dd HH:mm:ss"))
    .withColumn("level", F.trim(F.col("parts")[1]))
    .withColumn("message", F.trim(F.col("parts")[2]))
    .withColumn("server", F.trim(F.col("parts")[3]))
    .drop("parts", "raw_line") # keep only the clean columns
    .filter(F.col("timestamp").isNotNull()) # skip malformed lines
)

# Let's create a summary
summary_df = (
    parsed_df
    .groupBy("server", "level")
    .count()
    .orderBy("server", "level")
)

# Clean checkpoint
checkpoint_path = "/opt/spark/work-dir/checkpoints/logs_checkpoint"
dir_path = Path(checkpoint_path)
if dir_path.exists() and dir_path.is_dir():
    shutil.rmtree(dir_path)

# Write stream in the destination
query_events = (
    parsed_df.writeStream
    .outputMode("append") # append: show new rows only
    .format("console")
    .option("truncate", False) # don't cut off long messages
    .option("numRows", 20)
    .option("checkpointLocation", checkpoint_path)
    .queryName("parsed_logs")
    .start()
)
```

```
query_summary = (  
    summary_df.writeStream  
        .outputMode("complete")  
        .format("console")  
        .option("truncate", False)  
        .queryName("summary_logs")  
        .start()  
)  
  
print("    Press Ctrl+C to stop.\n")  
su.spark.streams.awaitAnyTermination()
```

26/04/07 02:46:01 WARN ResolveWriteToStream: spark.sql.adaptive.enabled is not supported in streaming DataFrames/Datasets and will be disabled.

26/04/07 02:46:01 WARN ResolveWriteToStream: Temporary checkpoint location created which is deleted normally when the query didn't fail: /tmp/temporary-b9f6ef7e-7d7e-4981-95bb-473af720d83d. If it's required to delete it under any circumstances, please set spark.sql.streaming.forceDeleteTempCheckpointLocation to true. Important to know deleting temp checkpoint folder is best effort.

26/04/07 02:46:01 WARN ResolveWriteToStream: spark.sql.adaptive.enabled is not supported in streaming DataFrames/Datasets and will be disabled.

Press Ctrl+C to stop.

Batch: 0

timestamp	level	message	server
2026-04-07 02:53:57	ERROR	Database connection timeout	server-node-1
2026-04-07 02:53:58	WARN	Certificate expires in 7 days	server-node-1
2026-04-07 02:54:00	INFO	Backup completed successfully	server-node-1
2026-04-07 02:54:06	WARN	Certificate expires in 7 days	server-node-4
2026-04-07 02:54:07	WARN	Memory usage 90%	server-node-3
2026-04-07 02:54:08	INFO	Configuration reloaded	server-node-1
2026-04-07 02:54:14	WARN	Memory usage 90%	server-node-2
2026-04-07 02:54:15	INFO	Cache cleared	server-node-5
2026-04-07 02:54:21	INFO	Health check passed	server-node-4
2026-04-07 02:54:24	ERROR	Authentication failed	server-node-2
2026-04-07 02:54:34	INFO	User login successful	server-node-2
2026-04-07 02:54:43	WARN	Retry attempt 3 of 5	server-node-4

Batch: 0

server	level	count
server-node-1	ERROR	1
server-node-1	INFO	2
server-node-1	WARN	1
server-node-2	ERROR	1
server-node-2	INFO	1
server-node-2	WARN	1
server-node-3	WARN	1
server-node-4	INFO	1
server-node-4	WARN	2
server-node-5	INFO	1

Batch: 1

timestamp	level	message	server
2026-04-07 02:54:17	INFO	Service restarted	server-node-2
2026-04-07 02:54:24	ERROR	Disk full	server-node-2
2026-04-07 02:54:25	ERROR	Unhandled exception in worker thread	server-node-4
2026-04-07 02:54:29	INFO	Backup completed successfully	server-node-3
2026-04-07 02:54:36	ERROR	Unhandled exception in worker thread	server-node-3
2026-04-07 02:54:41	WARN	Memory usage 90%	server-node-1
2026-04-07 02:54:47	ERROR	500 Internal Server Error	server-node-3
2026-04-07 02:54:48	INFO	Health check passed	server-node-5
2026-04-07 02:54:57	WARN	Disk usage 85%	server-node-5
2026-04-07 02:55:02	ERROR	404 Not Found	server-node-5
2026-04-07 02:55:11	INFO	User login successful	server-node-2
2026-04-07 02:55:17	ERROR	500 Internal Server Error	server-node-4

 Batch: 1

server	level	count
server-node-1	ERROR	1
server-node-1	INFO	2
server-node-1	WARN	2
server-node-2	ERROR	2
server-node-2	INFO	3
server-node-2	WARN	1
server-node-3	ERROR	2
server-node-3	INFO	1
server-node-3	WARN	1
server-node-4	ERROR	2
server-node-4	INFO	1
server-node-4	WARN	2
server-node-5	ERROR	1
server-node-5	INFO	2
server-node-5	WARN	1

 Batch: 2

timestamp	level	message	server
2026-04-07 02:54:37	ERROR	500 Internal Server Error	server-node-2
2026-04-07 02:54:45	INFO	Configuration reloaded	server-node-4
2026-04-07 02:54:50	WARN	Response time above threshold	server-node-4
2026-04-07 02:54:54	WARN	Disk usage 85%	server-node-2
2026-04-07 02:54:56	ERROR	Database connection timeout	server-node-1
2026-04-07 02:55:06	INFO	Health check passed	server-node-2
2026-04-07 02:55:11	ERROR	Disk full	server-node-4
2026-04-07 02:55:14	WARN	Retry attempt 3 of 5	server-node-2
2026-04-07 02:55:19	ERROR	404 Not Found	server-node-3
2026-04-07 02:55:26	INFO	Service restarted	server-node-4
2026-04-07 02:55:30	ERROR	Unhandled exception in worker thread	server-node-2
2026-04-07 02:55:38	WARN	Certificate expires in 7 days	server-node-3

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 Batch: 2

server	level	count
server-node-1	ERROR	2
server-node-1	INFO	2
server-node-1	WARN	2
server-node-2	ERROR	4
server-node-2	INFO	4
server-node-2	WARN	3
server-node-3	ERROR	3
server-node-3	INFO	1
server-node-3	WARN	2
server-node-4	ERROR	3
server-node-4	INFO	3
server-node-4	WARN	3
server-node-5	ERROR	1
server-node-5	INFO	2
server-node-5	WARN	1

 Batch: 3

timestamp	level	message	server
2026-04-07 02:54:57	INFO	Service restarted	server-node-2
2026-04-07 02:55:04	ERROR	404 Not Found	server-node-1
2026-04-07 02:55:12	ERROR	Disk full	server-node-3
2026-04-07 02:55:13	WARN	Response time above threshold	server-node-2
2026-04-07 02:55:17	INFO	Configuration reloaded	server-node-1
2026-04-07 02:55:19	INFO	Health check passed	server-node-5
2026-04-07 02:55:26	ERROR	Authentication failed	server-node-1
2026-04-07 02:55:34	INFO	Health check passed	server-node-1
2026-04-07 02:55:37	WARN	CPU spike detected	server-node-3
2026-04-07 02:55:41	WARN	Disk usage 85%	server-node-4
2026-04-07 02:55:49	INFO	Configuration reloaded	server-node-4
2026-04-07 02:55:59	INFO	Backup completed successfully	server-node-1

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Batch: 3

server	level	count
server-node-1	ERROR	4
server-node-1	INFO	5
server-node-1	WARN	2
server-node-2	ERROR	4
server-node-2	INFO	5
server-node-2	WARN	4
server-node-3	ERROR	4
server-node-3	INFO	1
server-node-3	WARN	3
server-node-4	ERROR	3
server-node-4	INFO	4
server-node-4	WARN	4
server-node-5	ERROR	1
server-node-5	INFO	3
server-node-5	WARN	1

Batch: 4

timestamp	level	message	server
2026-04-07 02:55:17	ERROR	500 Internal Server Error	server-node-2
2026-04-07 02:55:25	INFO	Configuration reloaded	server-node-2
2026-04-07 02:55:29	ERROR	Unhandled exception in worker thread	server-node-1
2026-04-07 02:55:33	INFO	Backup completed successfully	server-node-4
2026-04-07 02:55:42	INFO	Service restarted	server-node-3
2026-04-07 02:55:44	WARN	Certificate expires in 7 days	server-node-3
2026-04-07 02:55:53	WARN	Memory usage 90%	server-node-1
2026-04-07 02:56:00	WARN	Disk usage 85%	server-node-1
2026-04-07 02:56:03	INFO	Configuration reloaded	server-node-1
2026-04-07 02:56:10	INFO	Backup completed successfully	server-node-5
2026-04-07 02:56:15	WARN	Retry attempt 3 of 5	server-node-4
2026-04-07 02:56:16	ERROR	404 Not Found	server-node-3

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Batch: 4

```
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|server      |level|count|
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|server-node-1|ERROR|5     |
|server-node-1|INFO |6     |
|server-node-1|WARN |4     |
|server-node-2|ERROR|5     |
|server-node-2|INFO |6     |
|server-node-2|WARN |4     |
|server-node-3|ERROR|5     |
|server-node-3|INFO |2     |
|server-node-3|WARN |4     |
|server-node-4|ERROR|3     |
|server-node-4|INFO |5     |
|server-node-4|WARN |5     |
|server-node-5|ERROR|1     |
|server-node-5|INFO |4     |
|server-node-5|WARN |1     |
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```

In [5]: `su.spark.stop()`